Project Proposal

This project will investigate the use of wearable sensing to estimate human hand and wrist motion for intuitive real-time robotic teleoperation.

Project Outline

The background and motivation for the project

As the field of robotics continues to develop and robots become more common in industry and everyday life, the way people interact with and control them becomes increasingly important. Automation allows robots to carry out physical tasks with less direct human involvement, but some situations still benefit from human judgement and adaptability. This raises a question: how can we maintain a more present role in a robot's actions while benefiting from its physical capabilities? This is where I believe teleoperation has an important role. Through remote manipulation, a person can guide a robot's actions while remaining away from the task itself. For example, robotic systems can allow operators to handle hazardous materials from a safer location, reducing the need for direct exposure.

Although full automation may be suited for repetitive tasks in controlled environments, responding reliably to unfamiliar objects or unexpected changes can require more complex sensing and decision-making. Teleoperation allows the operator to assess the situation and adjust the robot's actions as the task develops. However, the operator also needs a suitable way to communicate those actions. Mapping human hand and wrist movements to robot commands offers a potentially more direct interface: closing the hand could command a gripper to close, while rotating the wrist could control the orientation of the robot's tool. The motivation is to investigate whether familiar human movements can provide a practical way to guide remote manipulation. This would allow the operator to express an intended action through movement rather than specify each robot joint individually.

The engineering challenge is that human movements must first be measured reliably and then translated into commands appropriate for the robot. A robot may have a different structure and range of movement from the human hand, so useful mapping requires deciding which aspects of the movements should be preserved. Sensor noise, differences in glove fit and calibration errors can affect the estimated motion, while filtering can improve signal smoothness at the cost of additional delay. These factors influence whether the robot responds consistently and promptly to the operator. This motivates my investigation into whether wearable measurements of hand and wrist movements can provide reliable commands for human-guided robotic manipulation, and how sensor selection, calibration and signal processing affect that response.

The overall project aim

The aim of this project is to develop and evaluate a wearable glove-based sensing system for real-time control, using the Ned2 as a primary test platform, using three human hand and wrist actions. Opening the hand will command the gripper to open, pointing will command the arm to extend within the defined workspace limits, and wrist movements will control the orientation of the robot's end effector.

The project will investigate how the sensor selection, calibration and signal processing influence the accuracy, repeatability and responsiveness of these commands. Experimental evaluation will assess how reliably the system detects the intended actions and how smoothly and promptly the robot responds.

The specific objectives

The specific objectives of this project are to:

1. Investigate and select suitable sensors: Review sensing technologies for detecting hand opening, pointing and wrist movement, and select suitable options based on measurement capability, cost and ease of integration into a wearable glove.
2. Characterise the selected sensors: Experimentally assess relevant characteristics including sensitivity, measurement range, linearity, hysteresis, repeatability, noise and drift, to identify limitations that could affect reliable control.
3. Develop the wearable glove: Design and construct a glove-based sensing system with appropriate sensor positioning, secure wiring and consideration of wearer comfort, safety and freedom of movement.
4. Calibrate the sensing system: Establish a relationship between sensor outputs and required hand and wrist actions, including thresholds for detecting hand opening and pointing, and a reference for measuring wrist orientation.
5. Develop signal processing methods: Implement and assess filtering and gesture detection methods to reduce noise and unintended commands while maintaining a responsive control signal
6. Implement human-to-robot mapping: Develop and integrate a control algorithm that maps hand opening to gripper, pointing to arm extension, and wrist movement to end-effector orientation on the Ned2 robotic arm, with appropriate movement limits and a stop mechanism.
7. Evaluate the complete system: Conduct repeatable tests to assess gesture detection accuracy, unintended activations, response latency, motion smoothness and repeatability, and identify how sensing and processing choices affect overall performance.

The initial scope of the project

The initial scope of this project is to investigate, develop and test a wearable glove-based sensing system for human-to-robot motion mapping, using the Ned2 robotic arm as the primary test platform. The work will focus on three control actions: opening the hand to open the gripper, pointing to command arm extension within defined limits, and moving the wrist to control the orientation of the robot's end effector. This will involve sensor selection and characterisation, glove construction, calibration, signal processing and integration with the robot.

Testing will take place in a controlled laboratory environment to assess how accurately, consistently and promptly the system responds to the intended actions. Pointing will act as a command to extend the arm in a predefined direction, rather than identifying an arbitrary target for the robot to reach. The project will focus on these selected actions rather than reproducing the complete movement of the human hand or developing autonomous manipulation. Although remote manipulation provides the motivation, deployment in a hazardous environment is outside the initial scope. Adaptation to other robotic arms or multi-fingered robotic hands could be explored as future development, depending on the results and time available.

Project Plan

Project Gantt Chart

Blue = task; teal = ongoing; M = milestone; Grey = Breal/exam period

Major task/Milestone	Start	Finish	Weeks	Type	28-Sep	05-Oct	12-Oct	19-Oct	26-Oct	02-Nov	09-Nov	16-Nov	23-Nov	30-Nov	07-Dec	14-Dec	Break	Break	Break	Break	Break	Break	01-Feb	08-Feb	15-Feb	22-Feb	01-Mar	08-Mar	15-Mar	22-Mar	29-Mar	05-Apr	12-Apr	19-Apr	26-Apr
Background research and requirements	28/09/2026	23/10/2026	4	Ongoing	x	x	x	x																											
Proposal, plan and risk assessments	12/10/2026	16/10/2026	1	Task			x																												
Proposal submission by 14:00	16/10/2026	16/10/2026	1	Milestone			M																												
Sensor selection and procurement	19/10/2026	30/10/2026	2	Task				x	x																										
Robot familiarisation and interface test	26/10/2026	06/11/2026	2	Task					x	x																									
Sensorcharacterisation	02/11/2026	20/11/2026	3	Task						x	x	x																							
Glove design and assembly	16/11/2026	04/12/2026	3	Task								x	x	x																					
Calibration and data logging	30/11/2026	11/12/2026	2	Task										x	x																				
Filtering and gesture detection	07/11/2026	18/12/2026	6	Task						x	x	x	x	x	x	x																			
Standalone glove prototype review	18/12/2026	18/12/2026	1	Milestone												M																			
Academic malpractice training by 14:00	18/12/2026	18/12/2026	1	Milestone												M																			
Robot mapping and controll intergration	01/02/2027	19/02/2027	3	Task																			x	x	x										
Intergrated system review	19/02/2027	19/02/2027	1	Milestone																					M										
Experimental evaluation	22/02/2027	12/03/2027	3	Task																						x	x	x							
Analysis and contingency testing	08/03/2027	19/03/2027	2	Task																								x	x						
Report development and draft preparation	19/10/2026	12/03/2027	21	Task				x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x							
Draft report submission by 14:00	12/03/2027	12/03/2027	1	Milestone																								M							
Final report and supervisor feedback	15/03/2027	16/04/2027	5	Task																									x	x	x	x	x		
Final report submission by 14:00	16/04/2027	16/04/2027	1	Milestone																													M		
Demonstration preparation and rehearsal	12/04/2027	23/04/2027	2	Task																													x	x	
Demonstration assessment week	26/04/2027	30/04/2027	1	Milestone																															M
Meetings, records and plan reviews	28/04/2027	30/04/2027	1	Task																															x

T1	Background research and requirements	None	Review wearable sensing and teleoperation approaches. Define the three commands, their priority, wrist axes and proposed performance measures with the supervisor.	Agreed requirements and initial test criteria
T2	Proposal, plan and risk assessments	T1 (initial findings)	Prepare the outline, provisional schedule, project risk register and health and safety assessment. Submit by the handbook deadline and obtain approval before practical work.	Submitted proposal and assessments
M1	Proposal submission by 14:00	T2	Handbook deadline: 16 October 2026 at 14:00. Submission does not itself constitute approval for practical work.	Proposal submitted
T3	Sensor selection and procurement	T1; budget and stock checks	Compare suitable sensors for the two hand gestures and wrist orientation. Confirm cost, availability and interfaces, then obtain the selected components.	Sensor choice justified and components available
T4	Robot familiarisation and interface test	H&S approval; robot access	Establish communication with the Ned2 and test a basic command under laboratory procedures. Confirm coordinate conventions, allowed movements and stopping arrangements before glove integration.	Basic robot command and stop tested
T5	Sensor characterisation	T3; H&S approval	Use repeatable bench tests to assess relevant range, sensitivity, hysteresis, repeatability, noise and drift. Record raw results and identify measurement limitations.	Characterisation dataset and analysis
T6	Glove design and assembly	T5 initial findings; H&S approval	Position and secure sensors and wiring in a wearable prototype. Check electrical insulation, fit, comfort and repeatability of placement before wearable trials.	Working glove with stable sensor readings
T7	Calibration and data logging	T6 usable prototype	Develop calibration for gesture thresholds and wrist reference orientation. Log timestamped measurements and repeat calibration trials to assess consistency.	Repeatable calibration and timestamped logs
T8	Filtering and gesture detection	T7 initial logs	Compare suitable filtering settings and define gesture detection, neutral states and command priorities. Evaluate noise reduction and processing delay using recorded data.	Glove commands detected without robot motion
M2	Standalone glove prototype review	T6–T8	Provisional internal milestone: demonstrate sensing, calibration and detection. Review remaining work with the supervisor before the inter-semester period.	Prototype readiness review
M3	Academic malpractice training by 14:00	None	Complete the handbook-required training and three related assessments by 18 December 2026 at 14:00.	Training and assessments completed
T9	Robot mapping and controlled integration	T4; T8; H&S approval; robot access	Integrate gripper opening, pointing-triggered extension and selected wrist orientation control one at a time. Implement movement limits, deliberate enable control and stopping on invalid or lost input.	Three actions demonstrated in restricted workspace
M4	Integrated system review	T9	Provisional internal milestone: verify the three mappings and stopping behaviour before formal performance trials. Confirm the experiment protocol with the supervisor.	System ready for evaluation
T10	Experimental evaluation	M4; approved test procedure	Run repeated trials of gesture detection and wrist control. Measure false activations, response latency, repeatability and motion smoothness using defined reference measurements.	Completed trials and organised raw data
T11	Analysis and contingency testing	T10 initial data	Analyse performance and measurement uncertainty. Reserve laboratory time for failed trials or limited corrections, then complete essential repeat tests before the break.	Final analysed results and limitations
T12	Report development and draft preparation	Research and results as available	Maintain methods, design notes, figures and references during active teaching weeks. Prepare a partly populated final-report draft with available results for the March deadline.	Partly populated report draft
M5	Draft report submission by 14:00	T12	Handbook deadline: 12 March 2027 at 14:00. Use supervisor feedback to agree the final structure and remaining revisions.	Draft report submitted
T13	Final report and supervisor feedback	T11; M5	Incorporate feedback and complete analysis, discussion and conclusions. Use available non-laboratory time over the break for writing if feasible, then check references, formatting and submission files.	Final report ready and submitted
M6	Final report submission by 14:00	T13	Handbook deadline: 16 April 2027 at 14:00. Plan an internal final check on 15 April rather than leaving upload until the deadline.	Final report submitted
T14	Demonstration preparation and rehearsal	Integrated system and results	Prepare the demonstration and explanation of results. Rehearse with the supervisor where possible and retain a recorded demonstration as a contingency, subject to assessment arrangements.	Rehearsed presentation and checked setup
M7	Demonstration assessment week	T14	Handbook assessment window is the week commencing 26 April 2027. The individual appointment date and time must be confirmed.	Final demonstration and presentation
T15	Meetings, records and plan reviews	None	Keep a project log, back up code and results, and review tasks, access and risks at supervisor meetings. Update dates and scope when progress or constraints change.	Current plan, risk register and project records

Weekly workload planner adapted from the supplied template

	S1 W1	S1 W2	S1 W3	S1 W4	S1 W5	S1 W6	S1 W7	S1 W8	S1 W9	S1 W10	S1 W11	S1 W12
	28-Sep	05-Oct	12-Oct	19-Oct	26-Oct	02-Nov	09-Nov	16-Nov	23-Nov	30-Nov	07-Dec	14-Dec
No. of deadlines (excl. project work)	0	0	1	2	0	1	3	0	1	2	1	2
Timetabled hours of on-campus activities (excl. project related activities)	10	10	10	10	15	12	12	12	13	15	15	13
Expected number of hours for private study (excl. project related activities)	15	15	15	15	15	15	15	15	15	15	15	15
Important Project Deadlines/Deliverables	0	0	Project Proposal and H&S Assessment, 16 Oct 2pm	0	0	0	0	0	0	0	0	Academic Malpractice, 18 Dec 2pm; M2
Expected number of hours for project work (incl. project meeting time)	10	10	10	10	10	10	10	10	10	10	10	10

	S2 W1	S2 W2	S2 W3	S2 W4	S2 W5	S2 W6	S2 W7	S2 W8	S2 W9	S2 W10	S2 W11	S2 W12
No. of deadlines (excl. project work)	01-Feb	08-Feb	15-Feb	22-Feb	01-Mar	08-Mar	15-Mar	22-Mar	29-Mar	05-Apr	12-Apr	19-Apr
Timetabled hours of on-campus activities (excl. project related activities)												
Expected number of hours for private study (excl. project related activities)												
Important Project Deadlines/Deliverables												
Expected number of hours for project work (incl. project meeting time)												